

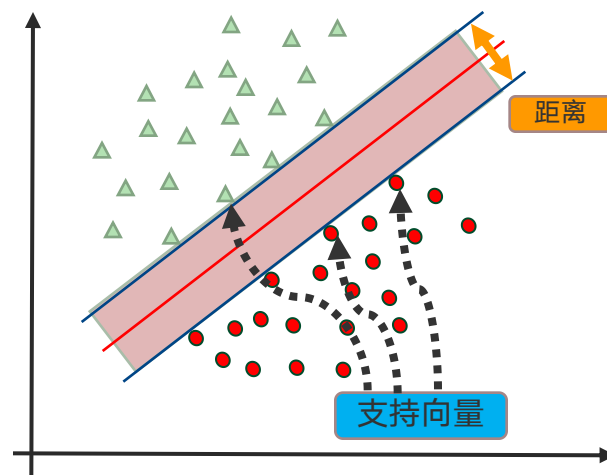
# 本讲内容

SVM分类

<https://cloud.tencent.com/developer/article/2318502?areaId=106001> (改)

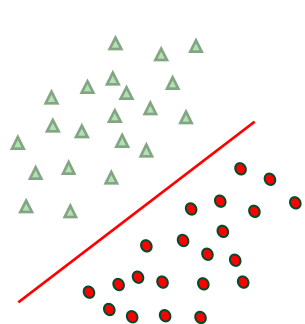
# 支持向量机概述

支持向量机（Support Vector Machine, SVM）是一类按有监督学习（supervised learning）方式对数据进行分类的广义线性分类器（generalized linear classifier），其决策边界是对学习样本求解的**最大边距超平面**（maximum-margin hyperplane）

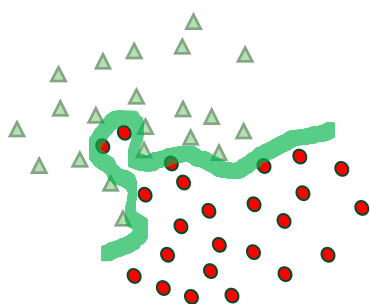


# 支持向量机概述

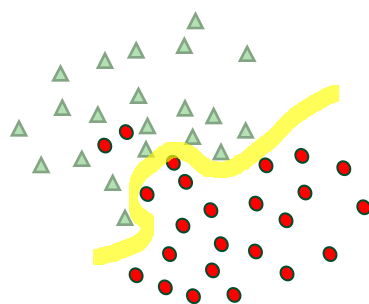
## 硬间隔、软间隔和非线性 SVM



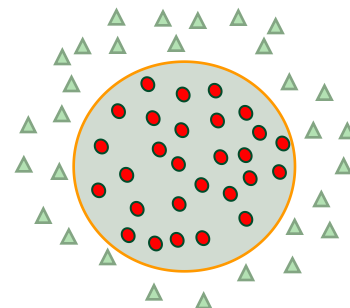
线性可分



硬间隔



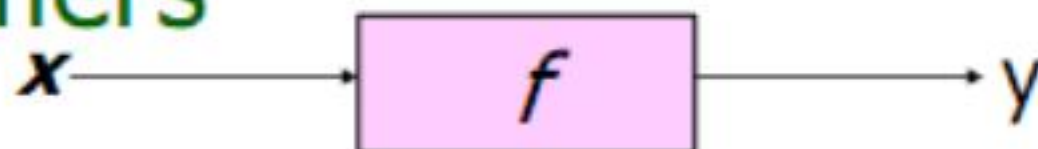
软间隔



线性不可分

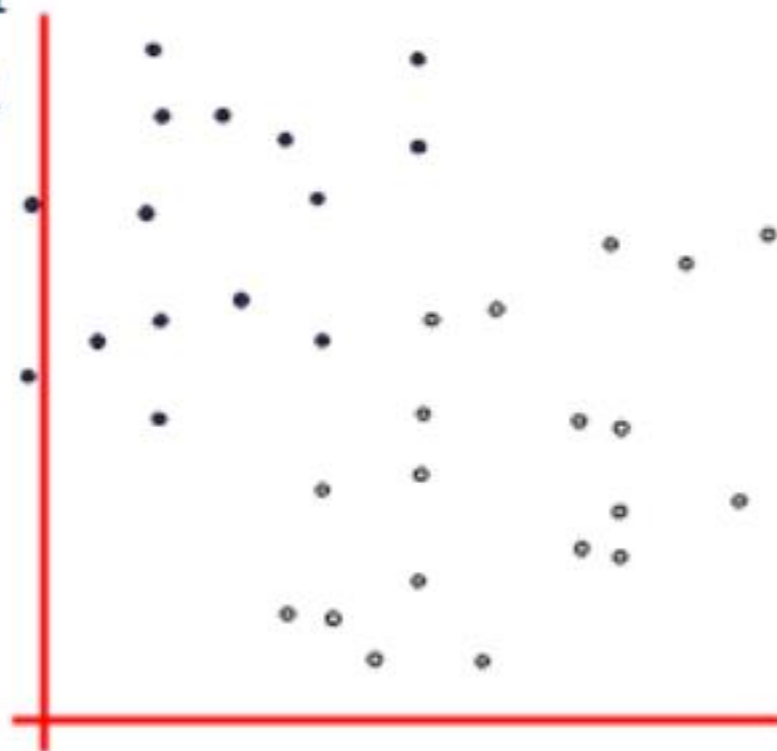
假如数据是完全的线性可分的，那么学习到的模型可以称为硬间隔支持向量机。换个说法，硬间隔指的就是完全分类准确，不能存在分类错误的情况。软间隔，就是允许一定量的样本分类错误。

# Linear Classifiers



$$f(x, w, b) = \text{sign}(w \cdot x + b)$$

- denotes +1
- denotes -1



How would you  
classify this data?

# Linear Classifiers

## (aka Linear Discriminant Functions)

- Definition

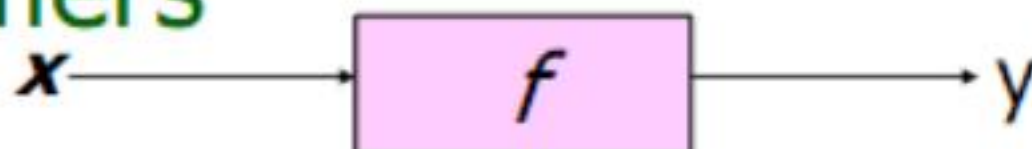
It is a function that is a linear combination of the components of the input  $\mathbf{x}$

$$f(\mathbf{x}) = \sum_{j=1}^m w_j x_j + b = \mathbf{w}^T \mathbf{x} + b$$

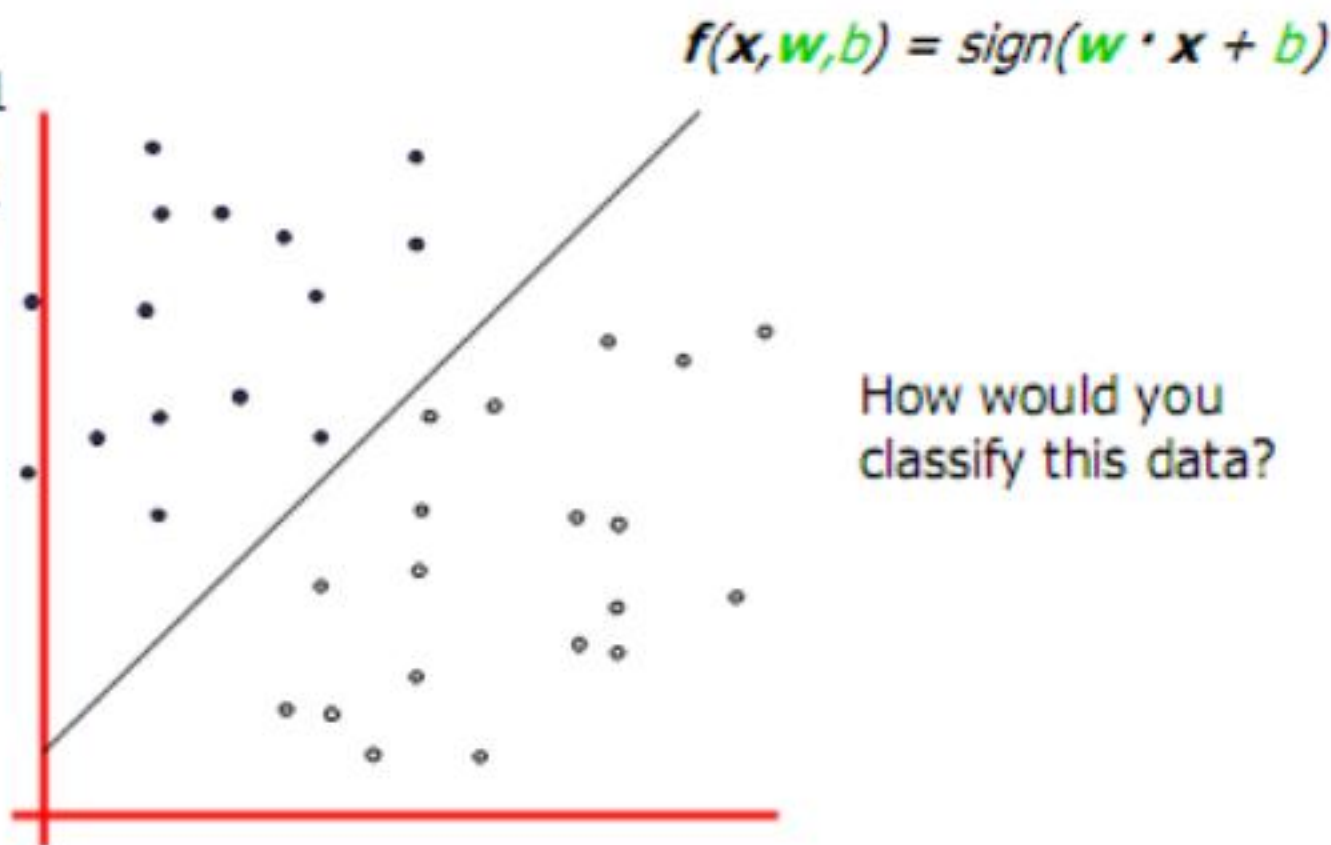
where  $\mathbf{w}$  is the weight vector and  $b$  the bias

- A two-category classifier then uses the rule:  
Decide class  $c_1$  if  $f(\mathbf{x}) > 0$  and class  $c_2$  if  $f(\mathbf{x}) < 0$   
 $\Leftrightarrow$  Decide  $c_1$  if  $\mathbf{w}^T \mathbf{x} > -b$  and  $c_2$  otherwise

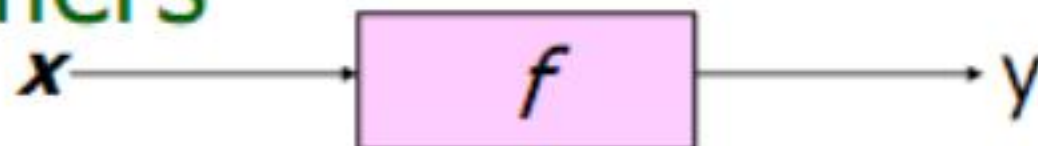
# Linear Classifiers



- denotes +1
- denotes -1

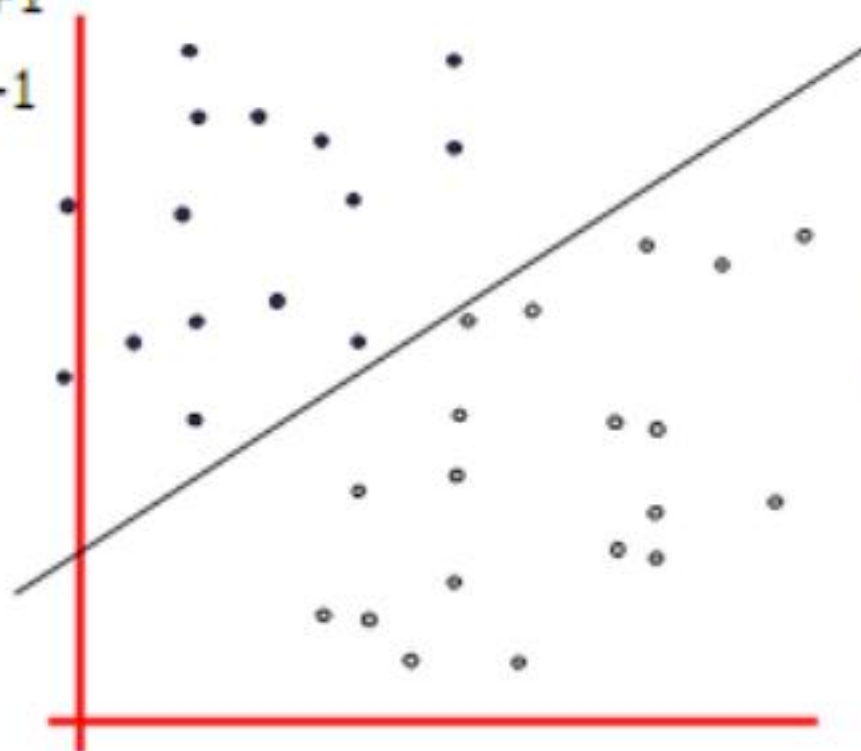


# Linear Classifiers



$$f(x, w, b) = \text{sign}(w \cdot x + b)$$

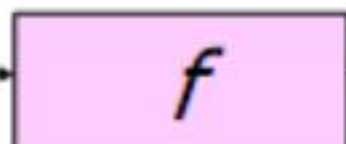
- denotes +1
- denotes -1



How would you  
classify this data?

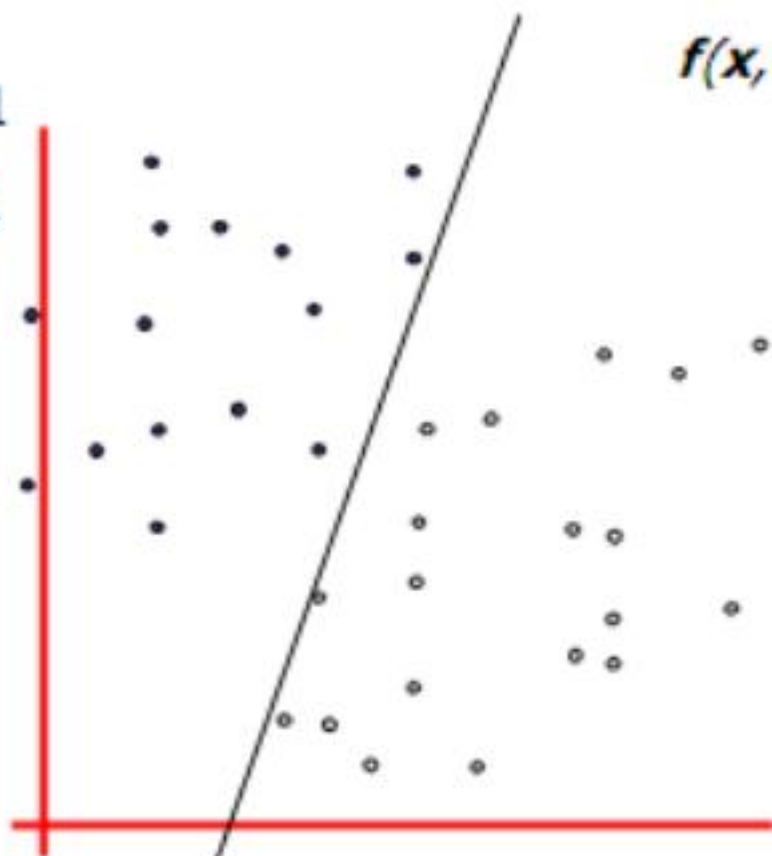
# Linear Classifiers

$\mathbf{x}$



$y$

- denotes +1
- denotes -1



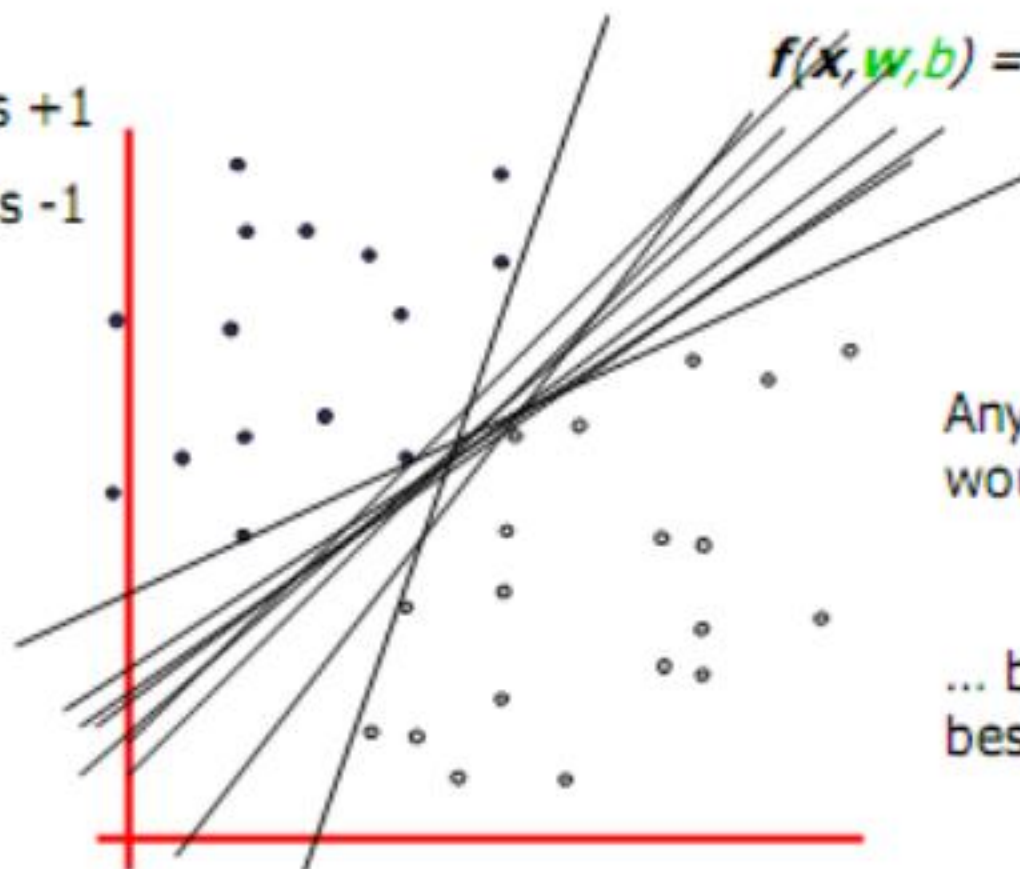
$$f(\mathbf{x}, \mathbf{w}, b) = \text{sign}(\mathbf{w} \cdot \mathbf{x} + b)$$

How would you  
classify this data?

# Linear Classifiers



- denotes +1
- denotes -1

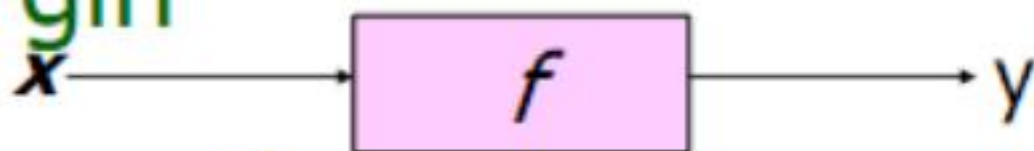


$$f(x, w, b) = \text{sign}(w \cdot x + b)$$

Any of these  
would be fine ...

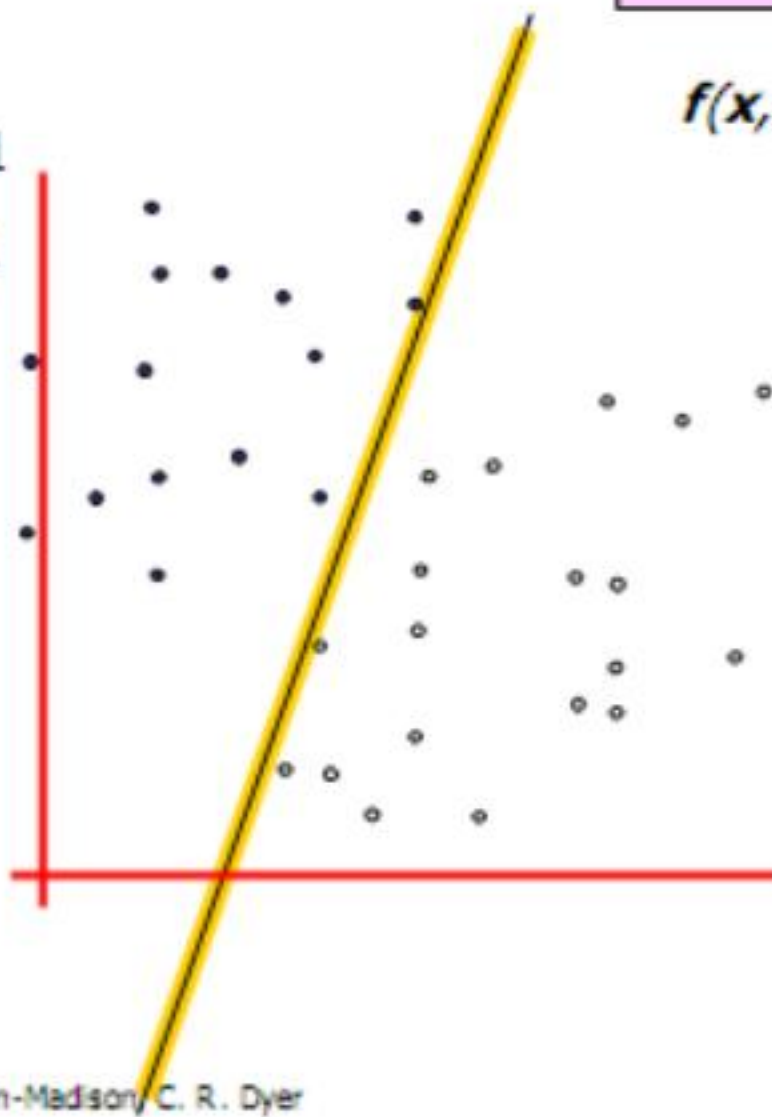
... but which is  
best?

# Classifier Margin



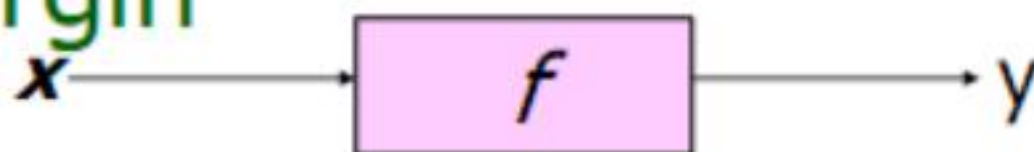
$$f(x, w, b) = \text{sign}(w \cdot x + b)$$

- denotes +1
- denotes -1

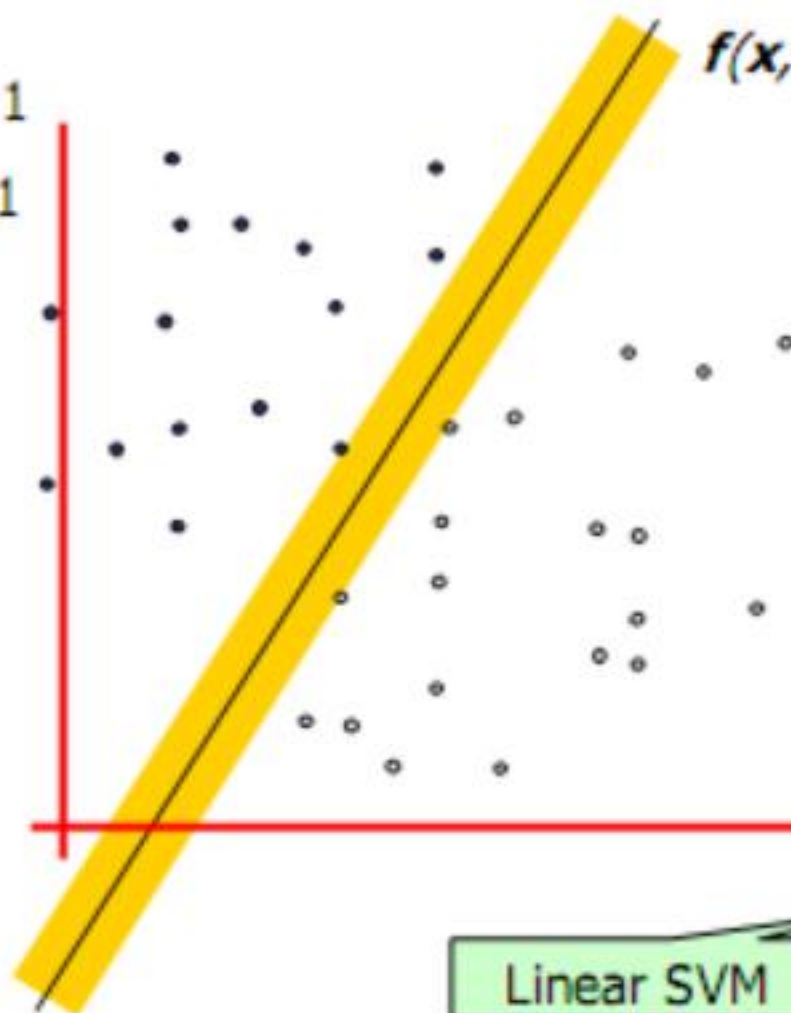


Define the **margin** of a linear classifier as the width that the boundary could be increased by before hitting a data point

# Maximum Margin



- denotes +1
- denotes -1



$$f(x, w, b) = \text{sign}(w \cdot x + b)$$

The **maximum margin linear classifier** is the linear classifier with the, um, maximum margin.

This is the simplest kind of SVM (Called an LSVM)

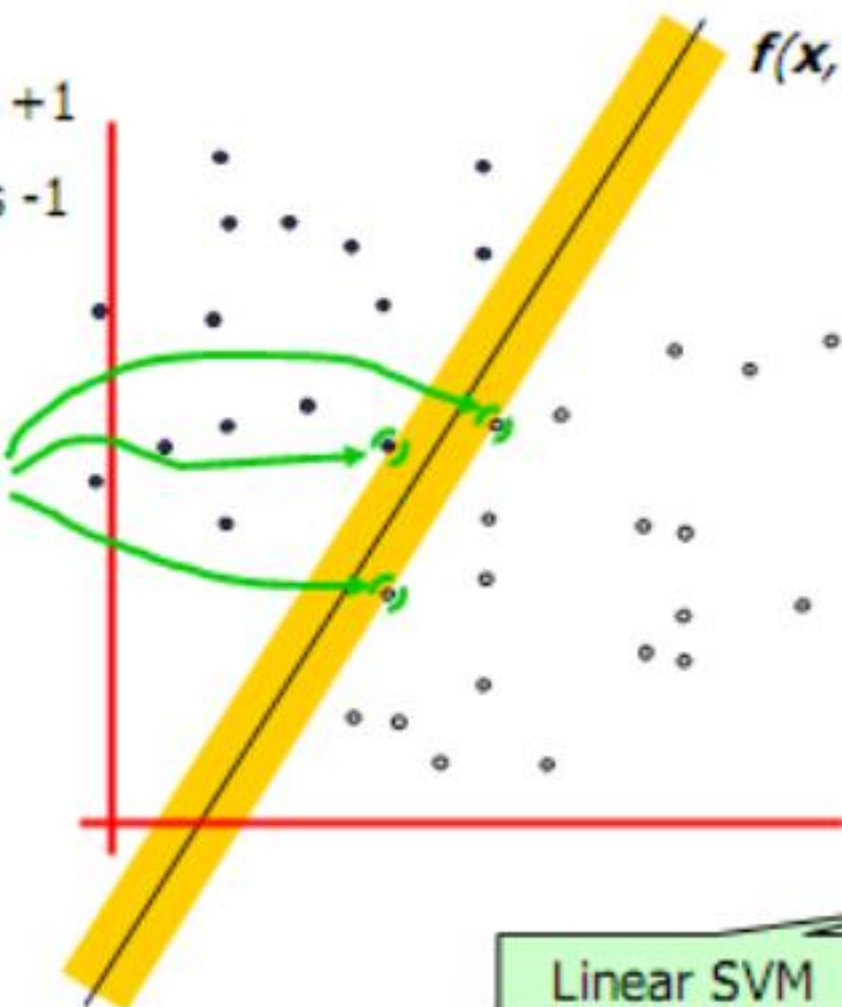
Linear SVM

# Maximum Margin



- denotes +1
- denotes -1

Support Vectors are those data points that the margin pushes up against



$$f(x, w, b) = \text{sign}(w \cdot x + b)$$

The maximum margin linear classifier is the linear classifier with the, um, maximum margin

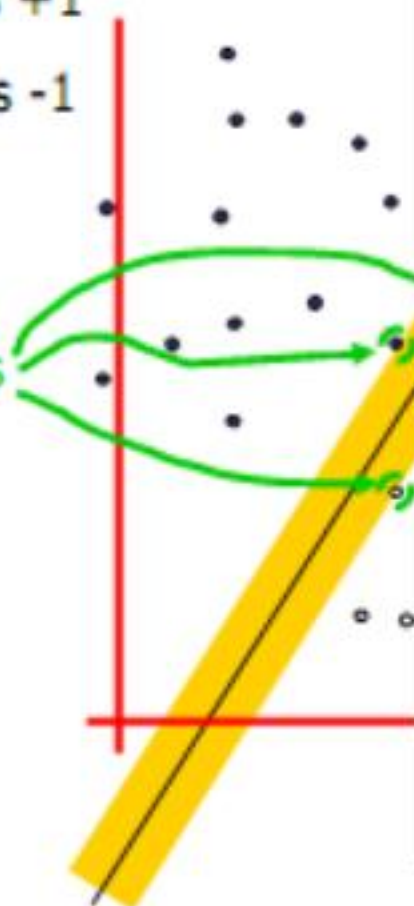
This is the simplest kind of SVM (Called an LSVM)

Linear SVM

# Why Maximum Margin?

- denotes +1
- denotes -1

Support Vectors are those data points that the margin pushes up against



1. Intuitively this feels safest
2. If we've made a small error in the location of the boundary (it's been jolted in its perpendicular direction) this gives us least chance of causing a misclassification
3. Robust to outliers since the model is immune to change/removal of any non-support-vector data points
4. There's some theory (using VC dimension) that is related to (but not the same as) the proposition that this is a good thing
5. Empirically it works very well

# 背景知识

- 任意超平面可以用下面这个线性方程来描述：

$$w^T x + b = 0$$

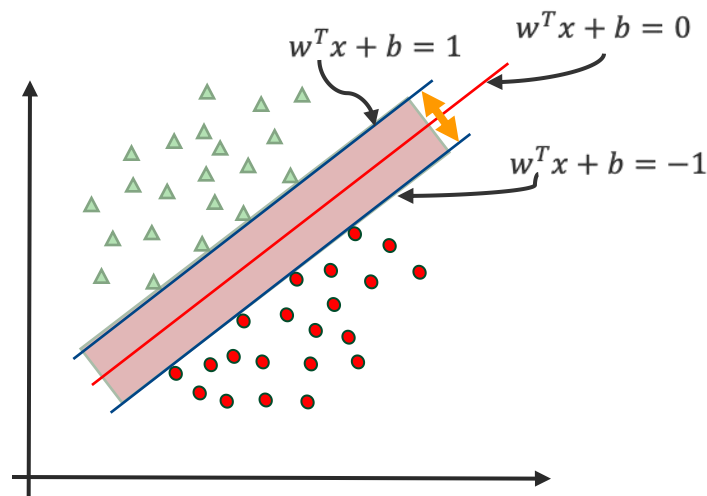
- 二维空间点  $(x, y)$  到直线  $Ax + By + C = 0$  的距离公式是：

$$\frac{|Ax + By + C|}{\sqrt{A^2 + B^2}}$$

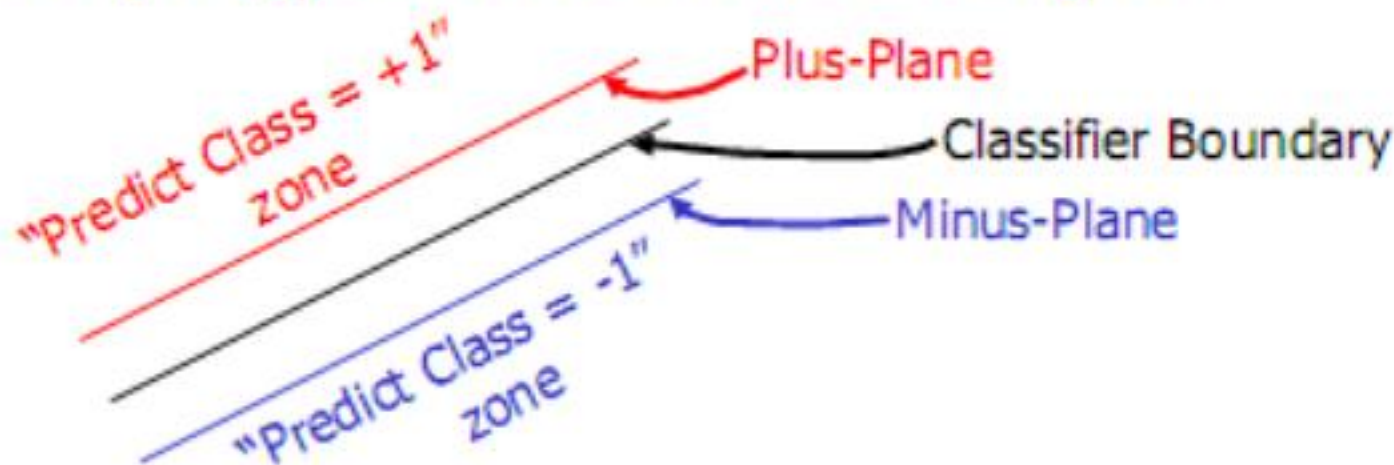
- 扩展到  $n$  维空间后，点  $x = (x_1, x_2 \dots x_n)$  到超平面

$$w^T x + b = 0 \text{ 的距离为: } \frac{|w^T x + b|}{\|w\|}$$

$$\text{其中 } \|w\| = \sqrt{w_1^2 + \dots w_n^2}$$

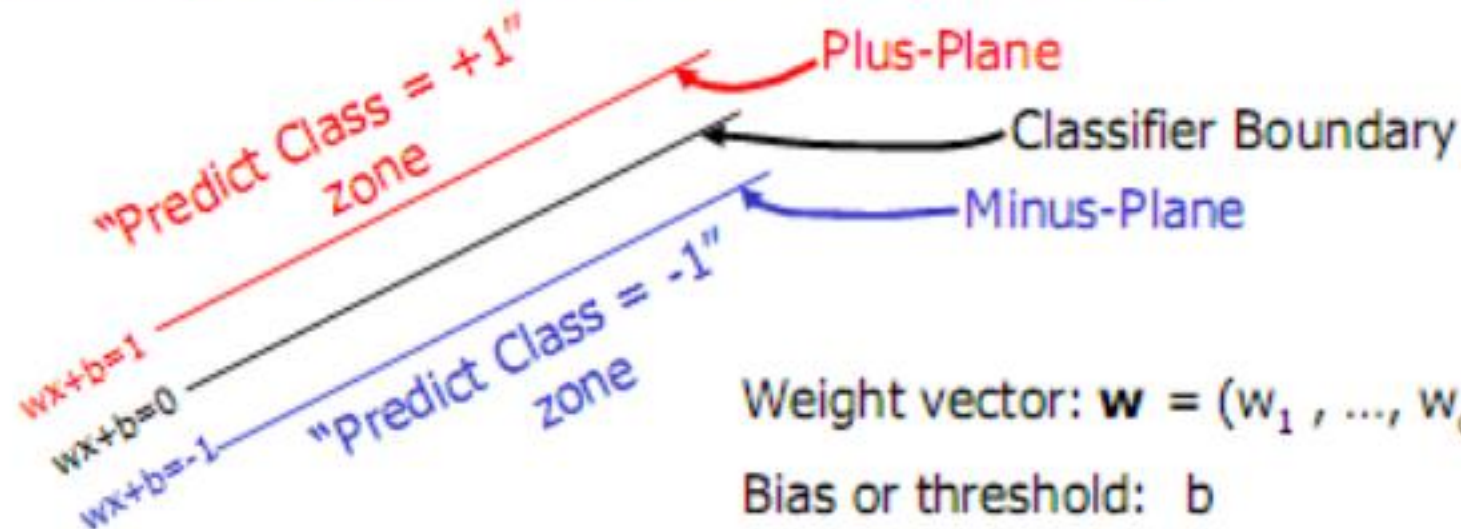


# Specifying a Line and Margin



- How do we represent this mathematically?
- ... in  $d$  input dimensions?
- An example  $\mathbf{x} = (x_1, \dots, x_d)^T$

# Specifying a Line and Margin



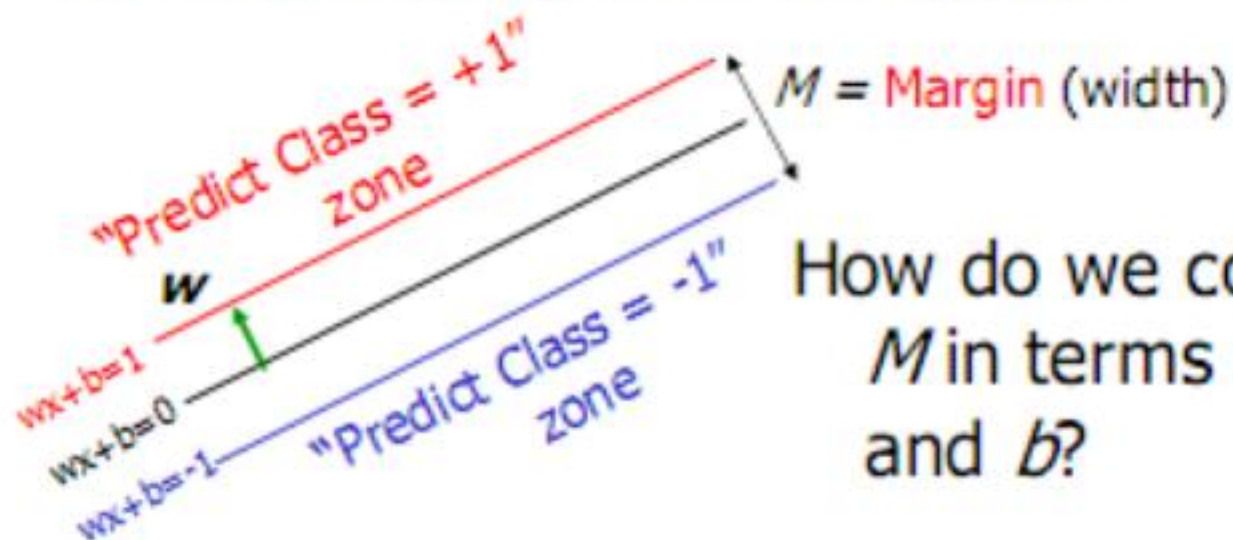
- Plus-plane =  $\{ \mathbf{w}^T \cdot \mathbf{x} + b = +1 \}$
- Minus-plane =  $\{ \mathbf{w}^T \cdot \mathbf{x} + b = -1 \}$

Classify as..  $+1$  if  $\mathbf{w}^T \cdot \mathbf{x} + b \geq 1$

$-1$  if  $\mathbf{w}^T \cdot \mathbf{x} + b \leq -1$

Universe explodes if  $-1 < \mathbf{w}^T \cdot \mathbf{x} + b < 1$

# Computing the Margin



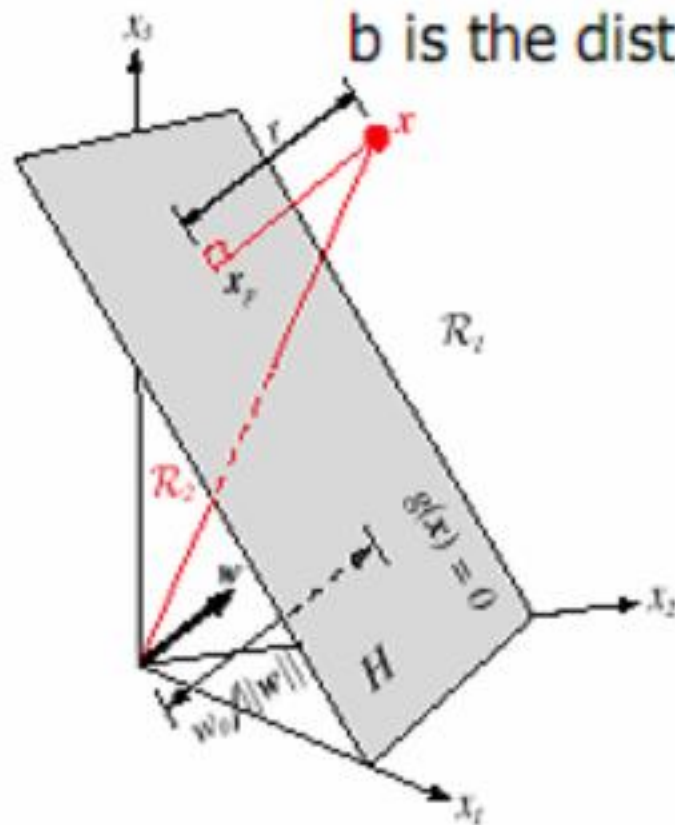
How do we compute  $M$  in terms of  $w$  and  $b$ ?

- Plus-plane =  $\{ w \cdot x + b = +1 \}$
- Minus-plane =  $\{ w \cdot x + b = -1 \}$

**Claim:** The vector  $w$  is perpendicular to the Plus-Plane

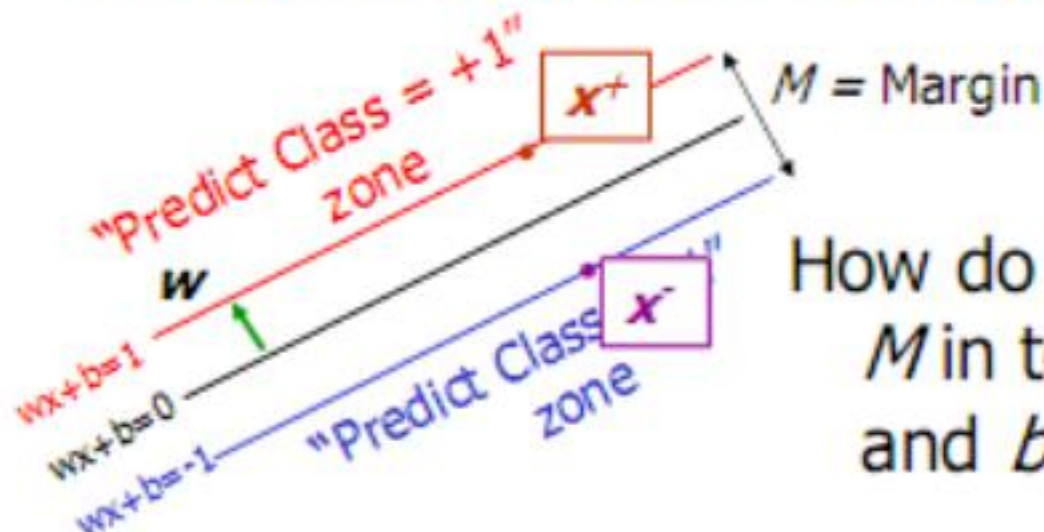
$\mathbf{w}$  is the plane's normal vector

$b$  is the distance from the origin



**FIGURE 5.2.** The linear decision boundary  $H$ , where  $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 = 0$ , separates the feature space into two half-spaces  $\mathcal{R}_1$  (where  $g(\mathbf{x}) > 0$ ) and  $\mathcal{R}_2$  (where  $g(\mathbf{x}) < 0$ ). From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

# Computing the Margin

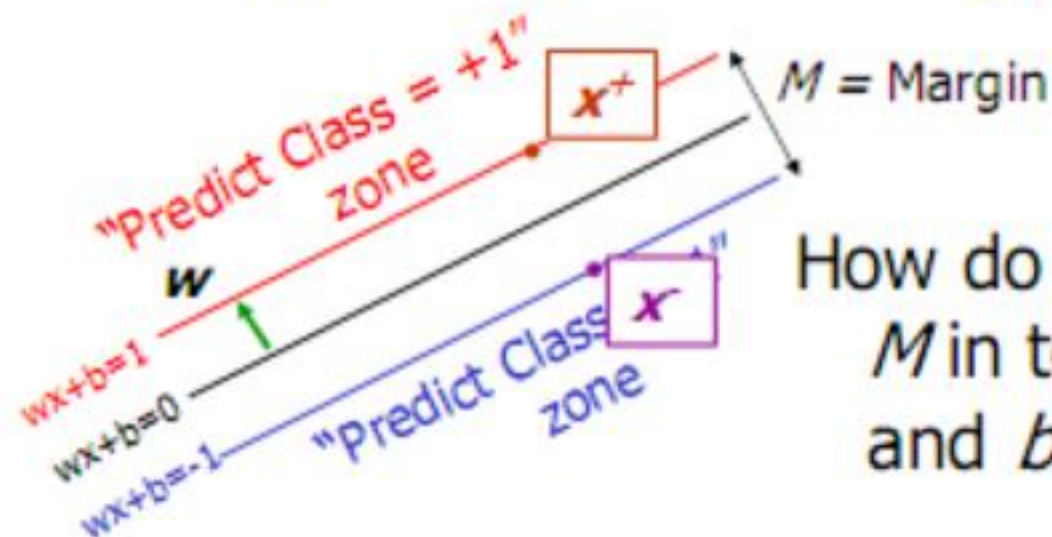


How do we compute  $M$  in terms of  $w$  and  $b$ ?

- Plus-plane =  $\{ w \cdot x + b = +1 \}$
- Minus-plane =  $\{ w \cdot x + b = -1 \}$
- The vector  $w$  is perpendicular to the Plus Plane
- Let  $x^-$  be any point on the minus plane
- Let  $x^+$  be the closest plus-plane-point to  $x^-$

Any location in  $\mathbb{R}^m$ : not necessarily a data point

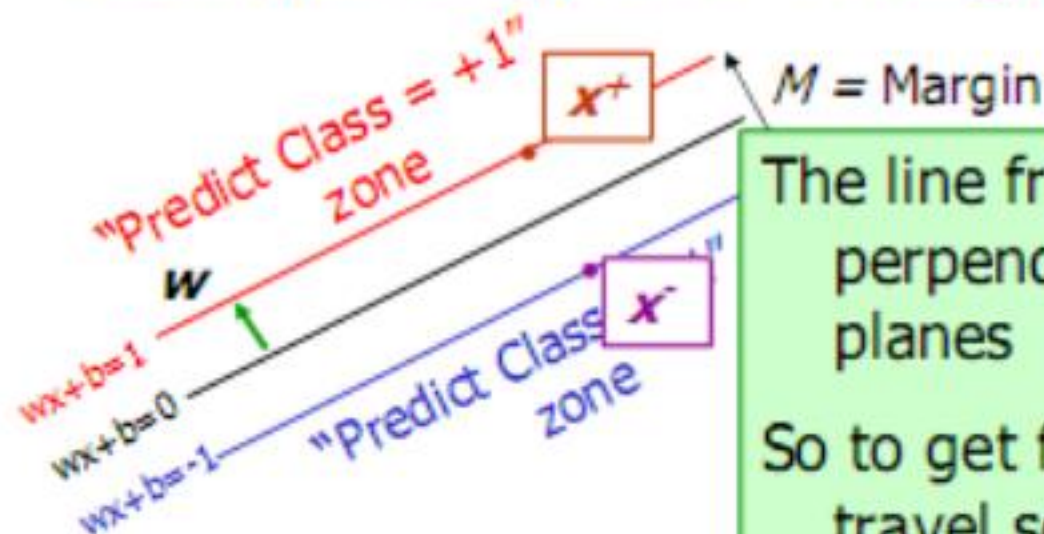
# Computing the Margin



How do we compute  $M$  in terms of  $w$  and  $b$ ?

- Plus-plane =  $\{ w \cdot x + b = +1 \}$
- Minus-plane =  $\{ w \cdot x + b = -1 \}$
- The vector  $w$  is perpendicular to the Plus Plane
- Let  $x$  be any point on the minus plane
- Let  $x^+$  be the closest plus-plane-point to  $x$
- **Claim:**  $x^+ = x + \lambda w$  for some value of  $\lambda$ . **Why?**

# Computing the Margin

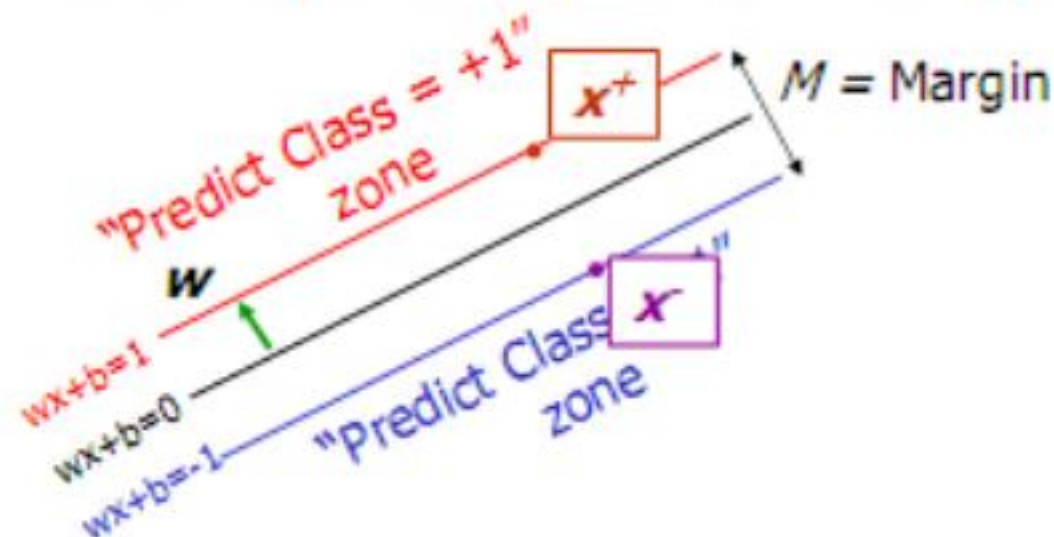


The line from  $x^-$  to  $x^+$  is perpendicular to the planes

So to get from  $x^-$  to  $x^+$  travel some distance in direction  $w$

- Plus-plane =  $\{ w \cdot x + b = +1 \}$
- Minus-plane =  $\{ w \cdot x + b = -1 \}$
- The vector  $w$  is perpendicular to the Plus Plane
- Let  $x^-$  be any point on the minus plane
- Let  $x^+$  be the closest plus-plane-point to  $x^-$
- **Claim:**  $x^+ = x^- + \lambda w$  for some value of  $\lambda$ . **Why?**

# Computing the Margin

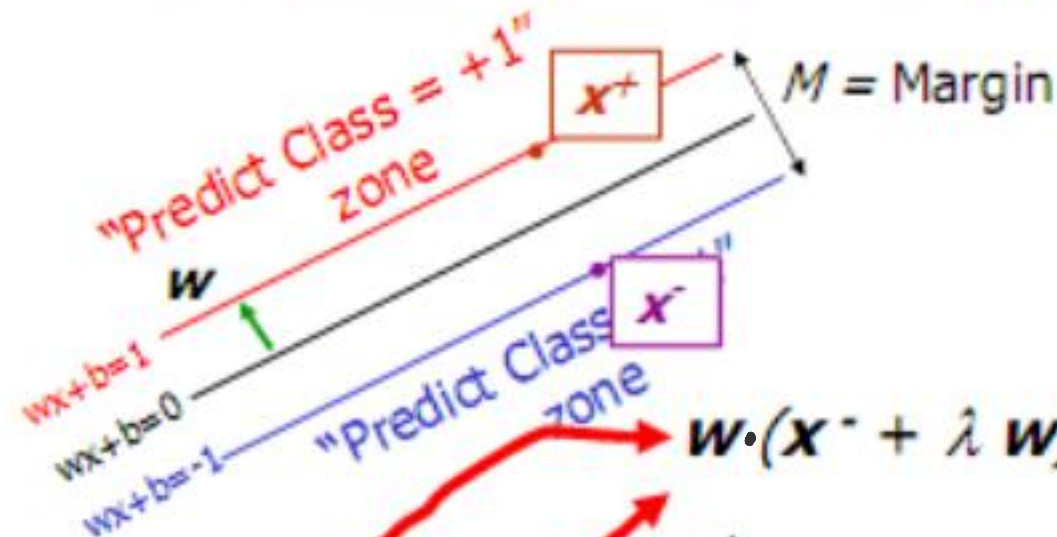


What we know:

- $w \cdot x^+ + b = +1$
- $w \cdot x + b = -1$
- $x^+ = x + \lambda w$
- $|x^+ - x| = M$

It's now easy to get  $M$  in terms of  $w$  and  $b$

# Computing the Margin



What we know:

- $w \cdot x^+ + b = +1$
- $w \cdot x^- + b = -1$
- $x^+ = x^- + \lambda w$
- $|x^+ - x^-| = M$

It's now easy to get  $M$   
in terms of  $w$  and  $b$

$$w \cdot (x^- + \lambda w) + b = 1$$

$\Rightarrow$

$$w \cdot x^- + b + \lambda w \cdot w = 1$$

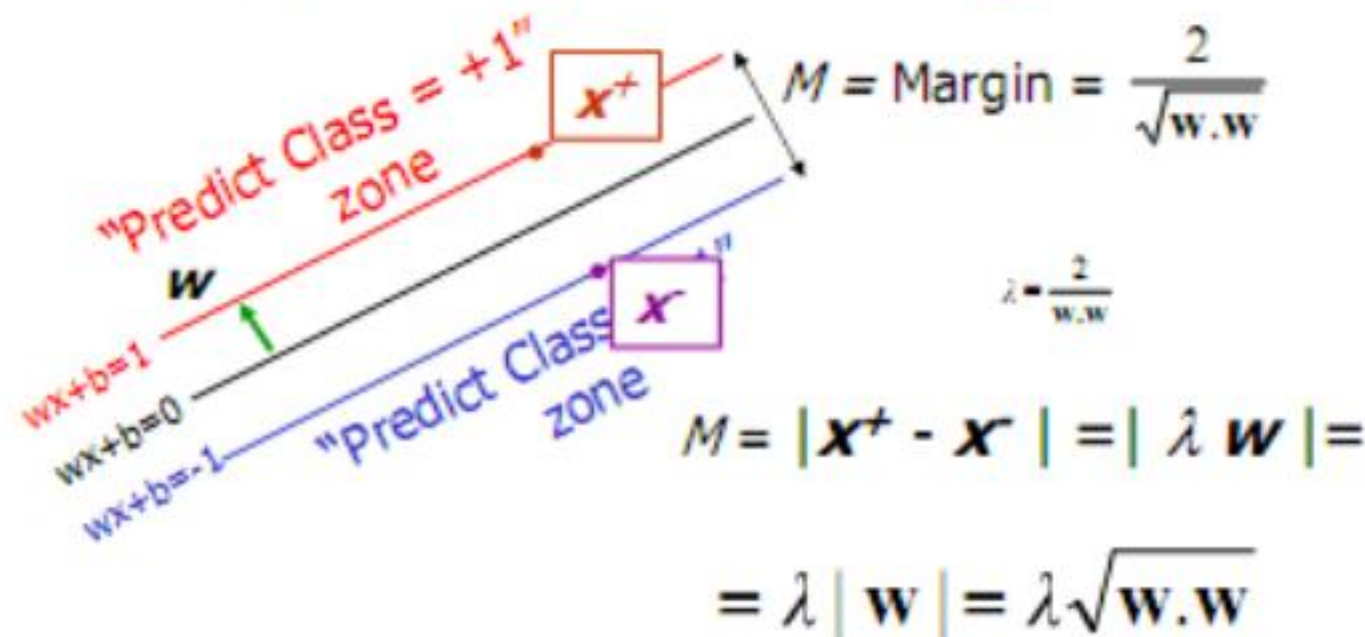
$\Rightarrow$

$$-1 + \lambda w \cdot w = 1$$

$\Rightarrow$

$$\lambda = \frac{2}{w \cdot w}$$

# Computing the Margin



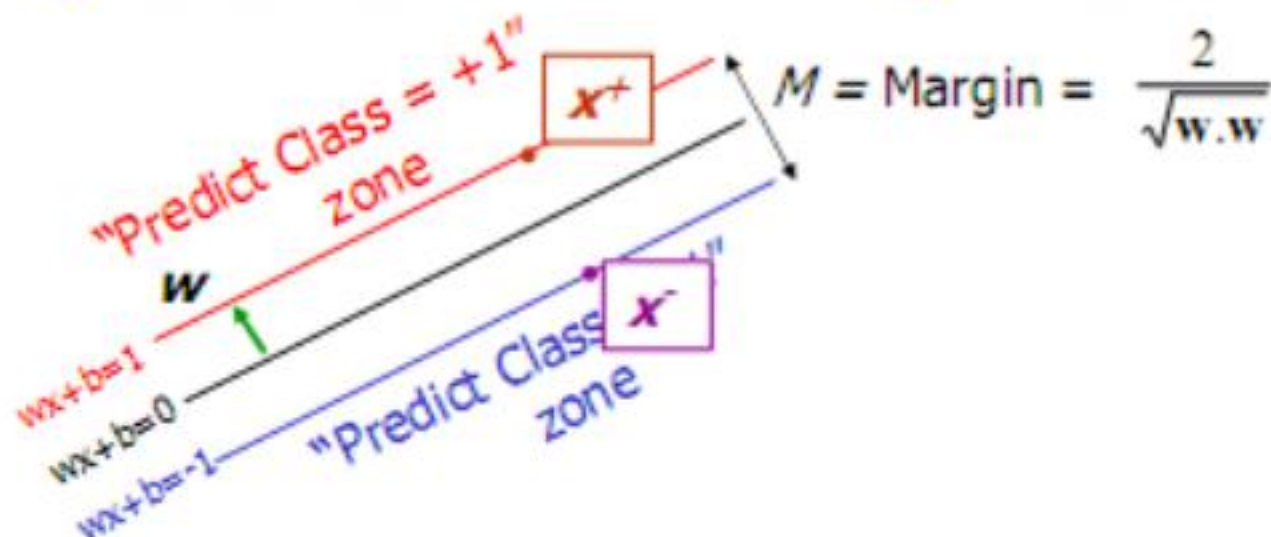
What we know:

- $w \cdot x^+ + b = +1$
- $w \cdot x^- + b = -1$
- $x^+ = x^- + \lambda w$
- $|x^+ - x^-| = M$

$$= \frac{2\sqrt{w \cdot w}}{w \cdot w} = \frac{2}{\sqrt{w \cdot w}}$$

$$= \frac{2}{\|w\|}$$

# Learning the Maximum Margin Classifier



Given a guess of  $w$  and  $b$  we can

- Compute whether all data points in the correct half-planes
- Compute the width of the margin

So now we just need to write a program to search the space of  $w$ 's and  $b$ 's to find the widest margin that matches all the data points. *How?*

# Learning via Quadratic Programming

- QP is a well-studied class of optimization algorithms to maximize a quadratic function of some real-valued variables subject to linear constraints
- Minimize  $\|\vec{w}\|^2$  subject to

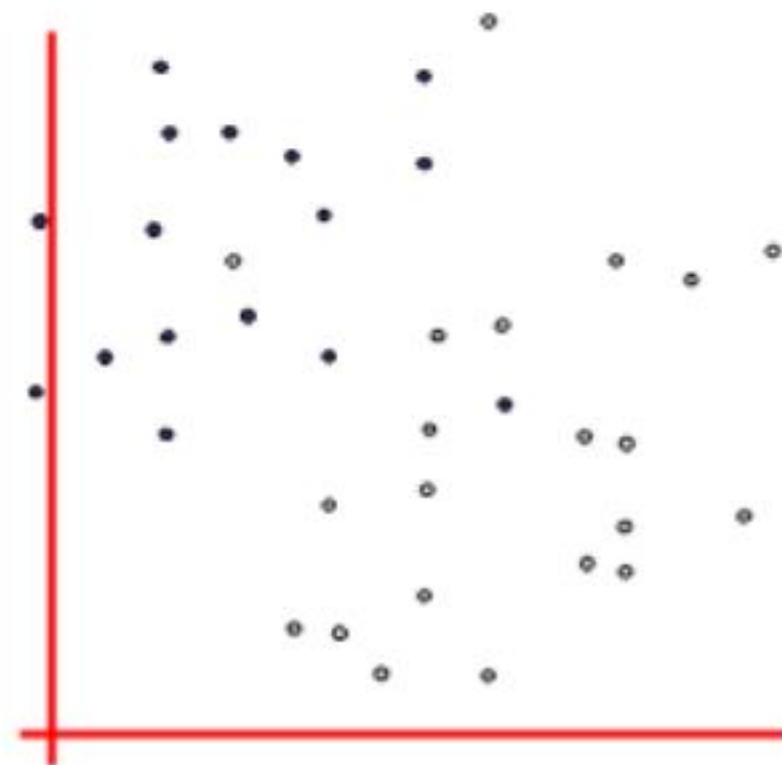
$$\mathbf{w} \cdot \mathbf{x} + b \geq +1 \quad \text{if } \mathbf{x} \text{ in class 1}$$

$$\mathbf{w} \cdot \mathbf{x} + b \leq -1 \quad \text{if } \mathbf{x} \text{ in class 2}$$

# Uh-oh!

This is going to be a problem!  
What should we do?

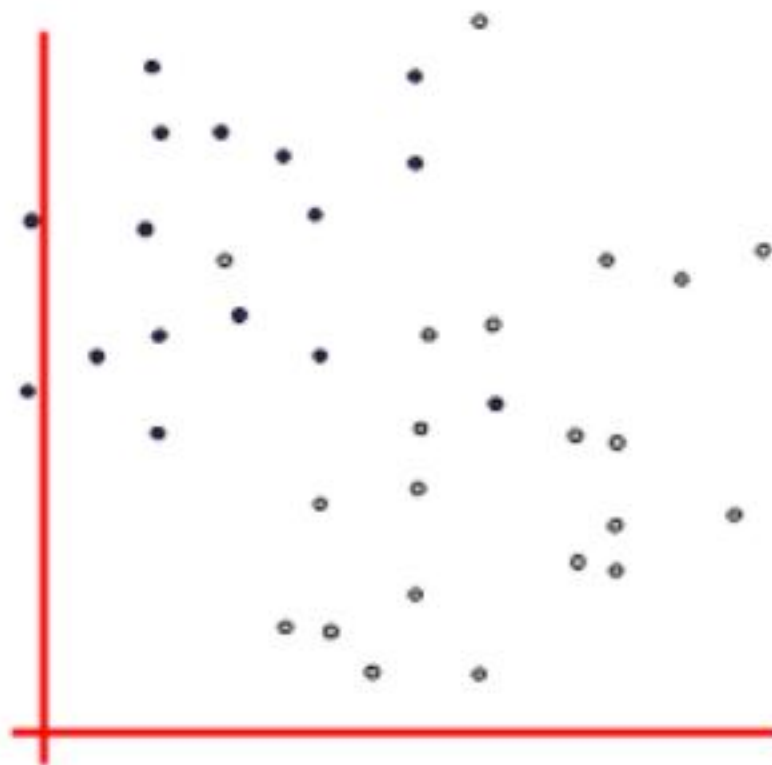
- denotes +1
- denotes -1



# Uh-oh!

This is going to be a problem!  
What should we do?

- denotes +1
- denotes -1



Idea 1:

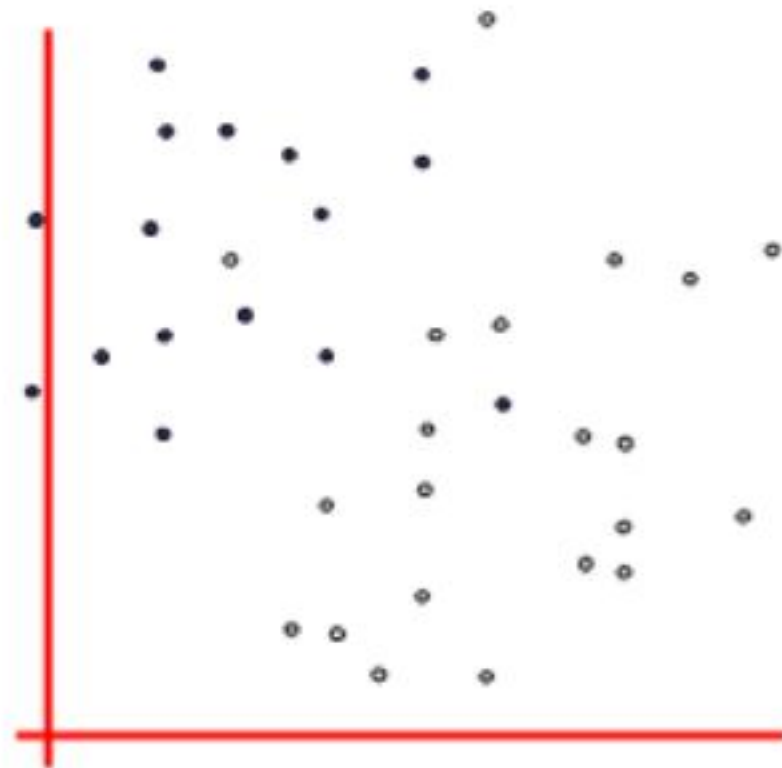
Find minimum  $||\mathbf{w}'||^2$ , while  
minimizing number of  
training set errors

Problem: Two things to  
minimize makes for an  
ill-defined optimization

# Uh-oh!

This is going to be a problem!  
What should we do?

- denotes +1
- denotes -1



Idea 1.1:

Minimize

$$||\mathbf{w}'||^2 + C (\#train\ errors)$$

Tradeoff parameter

There's a serious practical problem that's about to make us reject this approach. Can you guess what it is?

# Uh-oh!

This is going to be a problem!  
What should we do?

- denotes +1
- denotes -1

Idea 1.1:

Minimize

$$||w||^2 + C(\text{\#train errors})$$

Tradeoff parameter

Can't be expressed as a Quadratic Programming problem.

Solving it may be too slow.

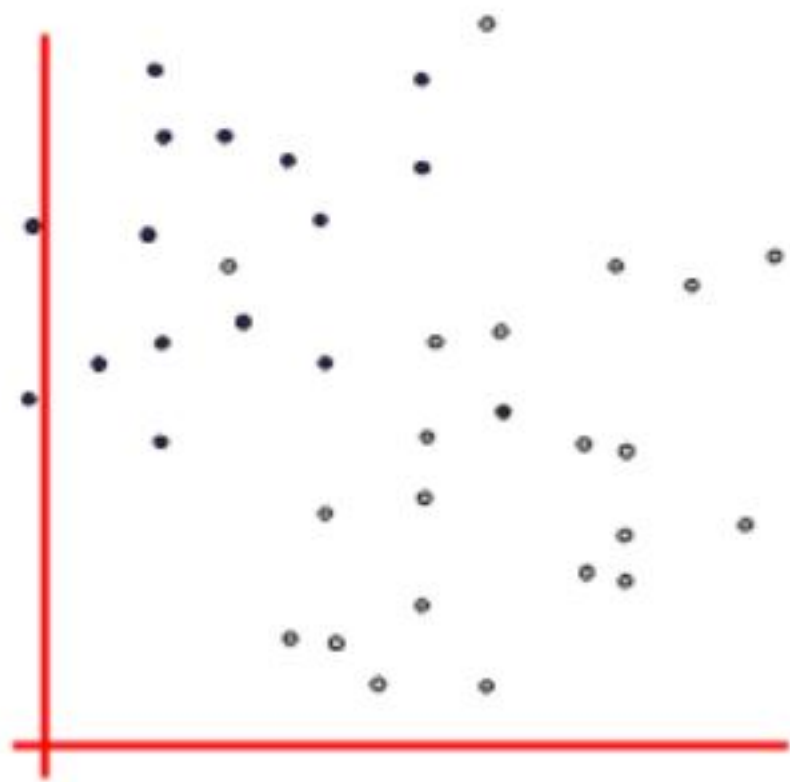
(Also, doesn't distinguish between disastrous errors and near misses)



# Uh-oh!

This is going to be a problem!  
What should we do?

- denotes +1
- denotes -1

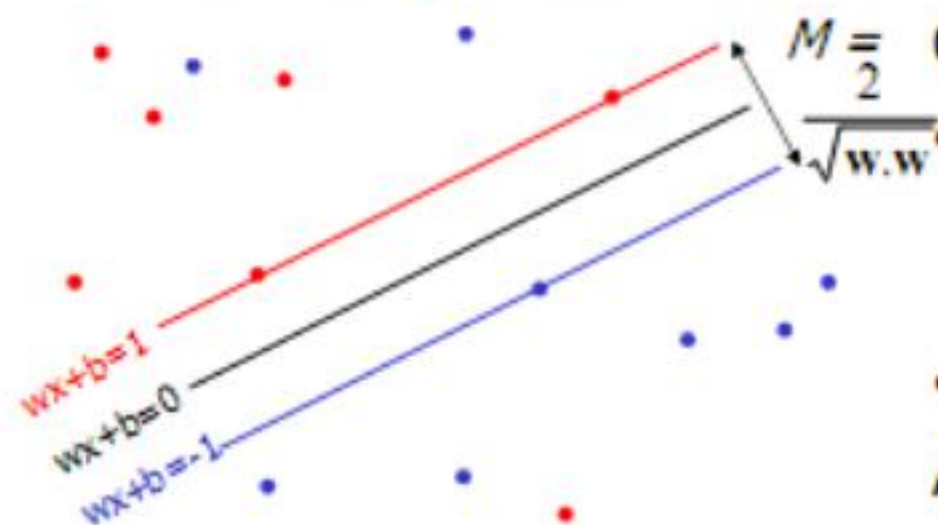


Idea 2.0:

Minimize

$||\mathbf{w}'||^2 + C$  (distance of  
error points to  
their correct  
place)

# Learning Maximum Margin with Noise



Given guess of  $w, b$ , we can

- Compute sum of distances of points to their correct zones

- Compute the margin width

Assume  $N$  examples, each

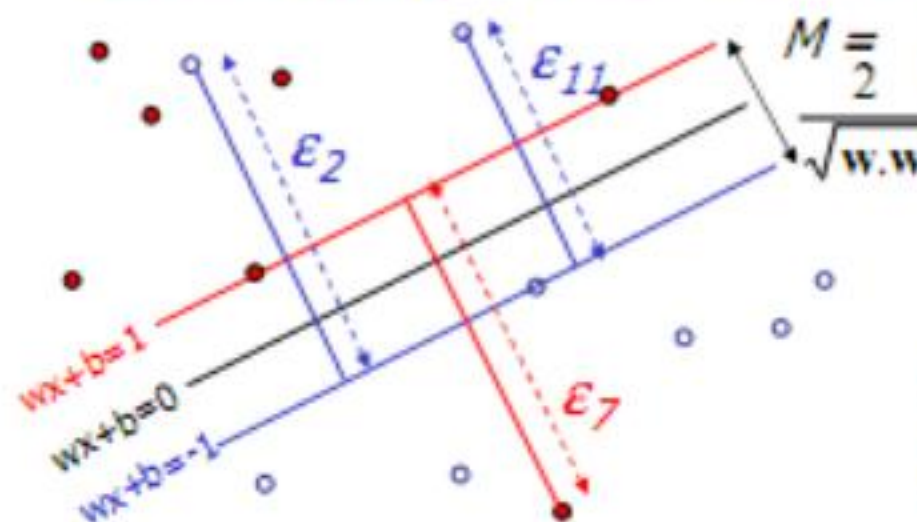
$(\mathbf{x}_k, y_k)$  where  $y_k = +/- 1$

What should our quadratic optimization criterion be?

How many constraints will we have?

What should they be?

# Learning Maximum Margin with Noise



- Given guess of  $\mathbf{w}$ ,  $b$  we can
- Compute sum of distances of points to their correct zones
  - Compute the margin width
- Assume  $N$  examples, each  $(\mathbf{x}_k, y_k)$  where  $y_k = +/- 1$

What should our quadratic optimization criterion be?

Minimize 
$$\frac{1}{2} \mathbf{w} \cdot \mathbf{w} + C \sum_{k=1}^N \epsilon_k$$

How many constraints will we have?  $N$

What should they be?

$\mathbf{w} \cdot \mathbf{x}_k + b \geq 1 - \epsilon_k$  if  $y_k = +1$

$\mathbf{w} \cdot \mathbf{x}_k + b \leq -1 + \epsilon_k$  if  $y_k = -1$

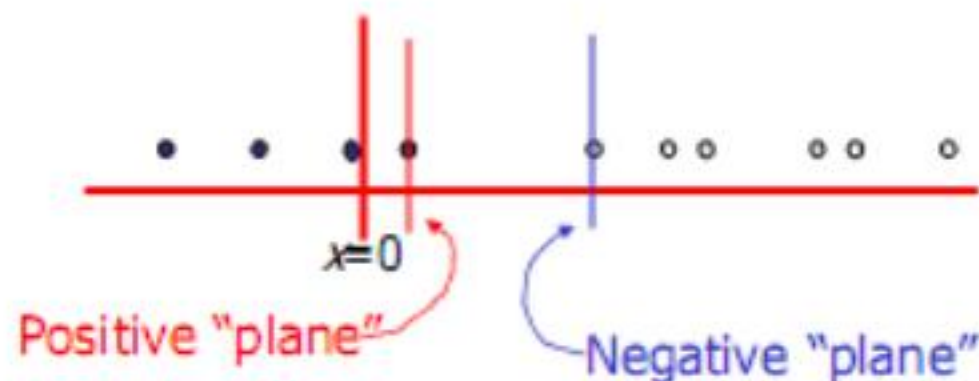
# Suppose we're in 1 Dimension

What would  
SVMs do with  
this data?



# Suppose we're in 1 Dimension

Not a big surprise



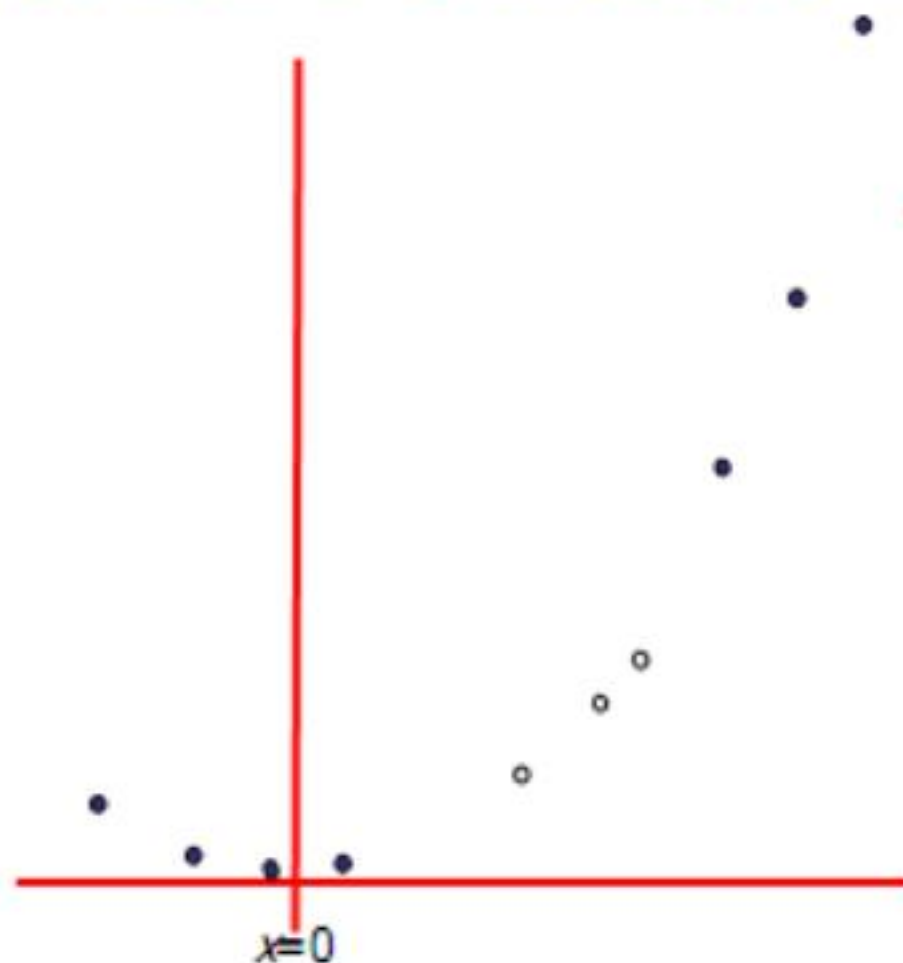
# Harder 1-Dimensional Dataset

That's wiped the  
smirk off SVM's  
face

What can be  
done about  
this?



# Harder 1-Dimensional Dataset

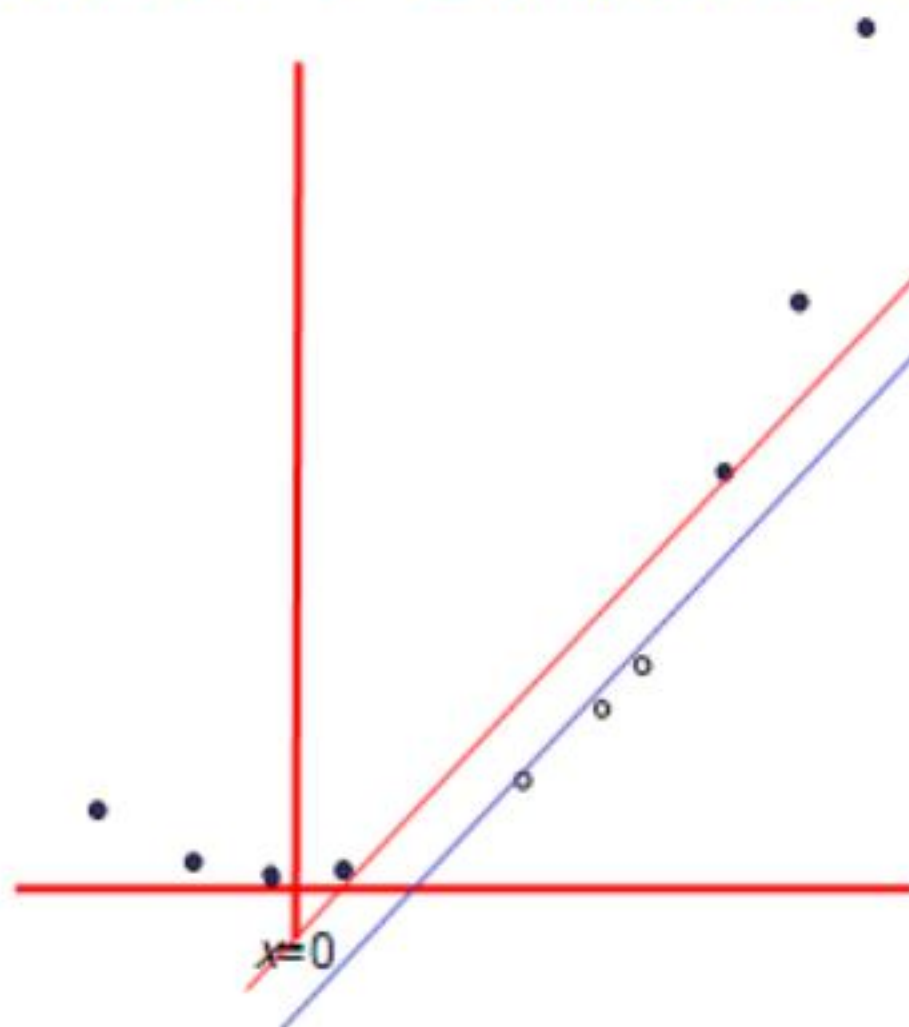


## The Kernel Trick:

Preprocess the data, mapping  $\mathbf{x}$  into higher dimensional space,  $F(\mathbf{x})$

$$\mathbf{z}_k = (x_k, x_k^2)$$

# Harder 1-Dimensional Dataset



## The Kernel Trick:

Preprocess the data, mapping  $\mathbf{x}$  into higher dimensional space,  $\mathcal{F}(\mathbf{x})$

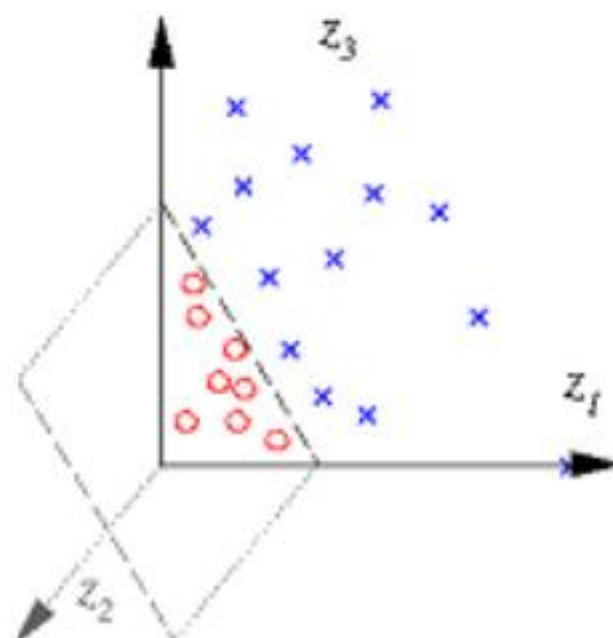
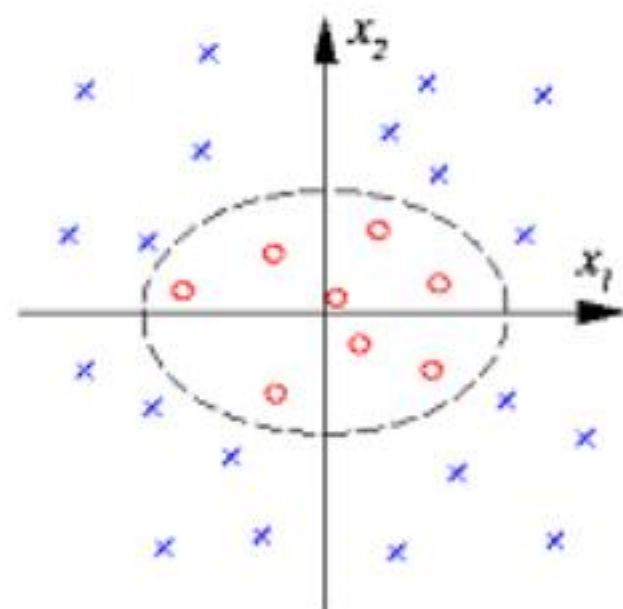
$$\mathbf{z}_k = (x_k, x_k^2)$$

## Example: All Degree 2 Monomials

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$$\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

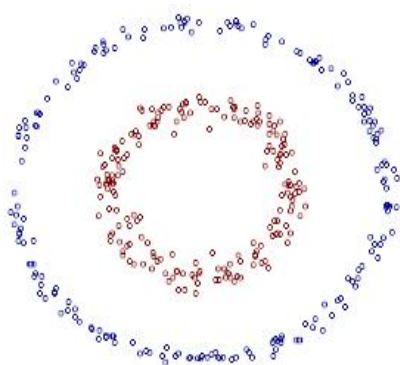
$$(x_1, x_2) \mapsto (z_1, z_2, z_3) := (x_1^2, \sqrt{2}x_1x_2, x_2^2)$$



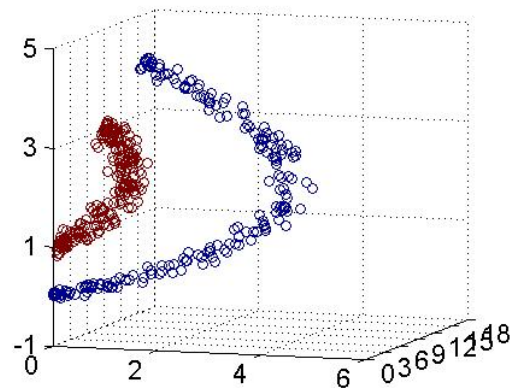
B. Schölkopf, NIPS, 3 December 2001

# 核技巧

在低维空间计算获得高维空间的计算结果，满足高维，才能在高维下线性可分。需要引入一个新的概念：**核函数**。它可以将样本从原始空间映射到一个更高维的特征空间中，使得样本在新的空间中线性可分。这样我们就可以使用原来的推导来进行计算，只是所有的推导是在新的空间中，而不是在原来的空间中进行。



线性不可分



高维下线性可分

# 核支持向量机

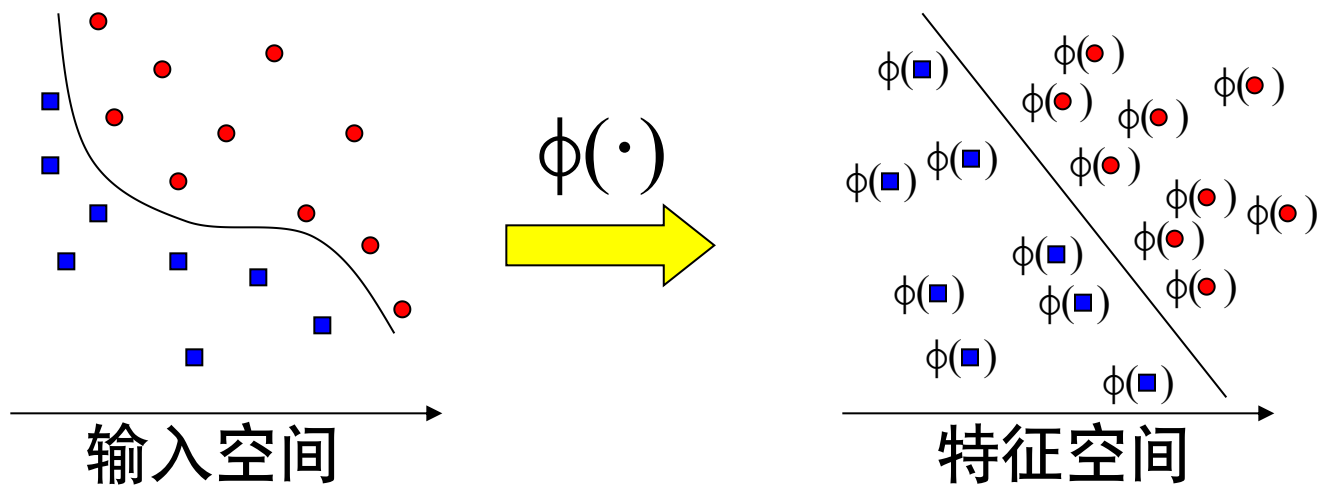
□ 设样本  $\mathbf{x}$  映射后的向量为  $\phi(\mathbf{x})$ , 划分超平面为  $f(\mathbf{x}) = \mathbf{w}^\top \phi(\mathbf{x}) + b$ .

原始问题

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{s.t. } y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1, i = 1, 2, \dots, m.$$

这里,  $y_i = +1$  或者  $-1$



# The Federalist Papers

- Written in 1787-1788 by Alexander Hamilton, John Jay, and James Madison to persuade the citizens of New York to ratify the constitution
- Papers consisted of short essays, 900 to 3500 words in length
- Authorship of 12 of those papers have been in dispute ( Madison or Hamilton); these papers are referred to as the disputed Federalist papers

# Description of the Data

- For every paper:
  - Machine readable text was created using a scanner
  - Computed relative frequencies of 70 words that Mosteller-Wallace identified as good candidates for author-attribution studies
  - Each document is represented as a vector containing the 70 real numbers corresponding to the 70 word frequencies
- The dataset consists of 118 papers:
  - 50 Madison papers
  - 56 Hamilton papers
  - 12 disputed papers

# Function Words Based on Relative Frequencies

|    |             |    |              |    |             |    |               |    |              |
|----|-------------|----|--------------|----|-------------|----|---------------|----|--------------|
| 1  | <i>a</i>    | 15 | <i>do</i>    | 29 | <i>is</i>   | 43 | <i>or</i>     | 57 | <i>this</i>  |
| 2  | <i>all</i>  | 16 | <i>down</i>  | 30 | <i>it</i>   | 44 | <i>our</i>    | 58 | <i>to</i>    |
| 3  | <i>also</i> | 17 | <i>even</i>  | 31 | <i>its</i>  | 45 | <i>shall</i>  | 59 | <i>up</i>    |
| 4  | <i>an</i>   | 18 | <i>every</i> | 32 | <i>may</i>  | 46 | <i>should</i> | 60 | <i>upon</i>  |
| 5  | <i>and</i>  | 19 | <i>for</i>   | 33 | <i>more</i> | 47 | <i>so</i>     | 61 | <i>was</i>   |
| 6  | <i>any</i>  | 20 | <i>from</i>  | 34 | <i>must</i> | 48 | <i>some</i>   | 62 | <i>were</i>  |
| 7  | <i>are</i>  | 21 | <i>had</i>   | 35 | <i>my</i>   | 49 | <i>such</i>   | 63 | <i>what</i>  |
| 8  | <i>as</i>   | 22 | <i>has</i>   | 36 | <i>no</i>   | 50 | <i>than</i>   | 64 | <i>when</i>  |
| 9  | <i>at</i>   | 23 | <i>have</i>  | 37 | <i>not</i>  | 51 | <i>that</i>   | 65 | <i>which</i> |
| 10 | <i>be</i>   | 24 | <i>her</i>   | 38 | <i>now</i>  | 52 | <i>the</i>    | 66 | <i>who</i>   |
| 11 | <i>been</i> | 25 | <i>his</i>   | 39 | <i>of</i>   | 53 | <i>their</i>  | 67 | <i>will</i>  |
| 12 | <i>but</i>  | 26 | <i>if</i>    | 40 | <i>on</i>   | 54 | <i>then</i>   | 68 | <i>with</i>  |
| 13 | <i>by</i>   | 27 | <i>in</i>    | 41 | <i>one</i>  | 55 | <i>there</i>  | 69 | <i>would</i> |
| 14 | <i>can</i>  | 28 | <i>into</i>  | 42 | <i>only</i> | 56 | <i>things</i> | 70 | <i>your</i>  |

# SLA Feature Selection for Classifying the Disputed Federalist Papers

- Apply the SVM Successive Linearization Algorithm for feature selection to:
  - Train on the 106 Federalist papers with known authors
  - Find a classification hyperplane that uses as few words as possible
- Use the hyperplane to classify the 12 disputed papers

# Hyperplane Classifier Using 3 Words

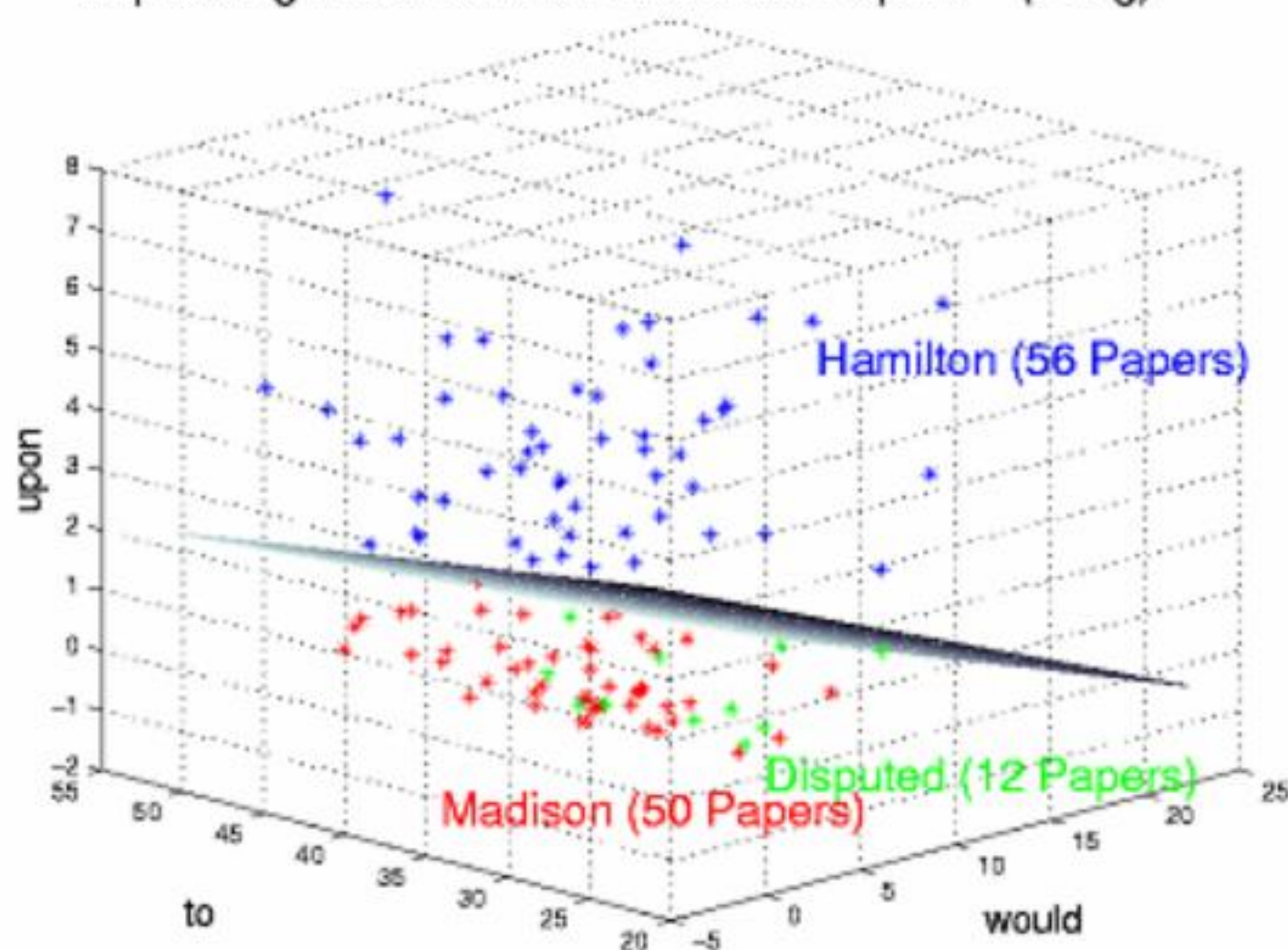
- A hyperplane depending on three words was found:

$$0.537\textit{to} + 24.663\textit{upon} + 2.953\textit{would} = 66.616$$

- All disputed papers ended up on the Madison side of the plane

# Results: 3D Plot of Hyperplane

Separating Plane for the Federalists Papers – (Fung)



# Multi-Class Classification

- SVMs can only handle two-class outputs
- What can be done?
- Answer: for N-class problems, learn N SVM's:
  - SVM 1,  $f_1$ , learns "Output=1" vs "Output  $\neq$  1"
  - SVM 2,  $f_2$ , learns "Output=2" vs "Output  $\neq$  2"
  - :
  - SVM N,  $f_N$ , learns "Output=N" vs "Output  $\neq$  N"

# Multi-Class Classification

- Ideally, only one  $f_i(\mathbf{x}) > 0$  and all others  $< 0$ , but this is not often the case in practice
- Instead, to predict the output for a new input, just predict with each SVM and find out which one puts the prediction the furthest into the positive region:
  - Classify as class  $C_i$  if  $f_i(\mathbf{x}) = \max \{ f_j(\mathbf{x}) \}$  for all  $j$

# Summary

- Learning linear functions
  - Pick separating plane that maximizes margin
  - Separating plane defined in terms of support vectors only
- Learning non-linear functions
  - Project examples into higher dimensional space
  - Use kernel functions for efficiency
- Generally avoids over-fitting problem
- Global optimization method; no local optima
- Can be expensive to apply, especially for multi-class problems